

BE 101: Reverse Engineering for Bioengineers

Course Code	BE 101
Course Name	Reverse Engineering for Bioengineers
Offered in	B.Tech Bioengineering (exit option from IDD B.Tech-M.Tech Bioengineering, 2021 & 2022 batches)
Approval	recommended by the 59th BoA (Apr 2025)

This course follows the syllabus of BE 101P, reproduced below.

Course Code	BE 101P
Course Name	Reverse Engineering for Bioengineers
L-T-P-C	0-0-2-1
Prerequisites	No prerequisite (compulsory for Bioengineering students)
Intended for	B.Tech.-M.Tech. Integrated Dual Degree in Bioengineering (Core Course)
Mutual Exclusion	NA
Approval	49th BoA (Mar 2023)

Course Modules

The students focus on hardware reverse engineering (RE). In the process of RE students understand existing technologies, functions, features, objects, components and systems. By carefully disassembling, observing, testing, analyzing and reporting, students can understand how something works and suggest ways it might be improved. This process requires careful observation, disassembly, documentation, analysis and reporting. Many times, the reverse engineering process is non-destructive. This means that the object or component can be reassembled and still function just as it did before it was taken apart. Throughout the reverse engineering project, the students are able to think of ways these objects could be improved. Is there some way it could function better? or manufactured less expensively? The students will use observations to make suggestions for improvement of the product. (28 Hours)

Learning Topics

Reverse Engineering of Biotechnological/Biomedical- Devices/ prototypes.

Text Books

1. Lam, R.H. and Chen, W., **Biomedical Devices: Materials, Design, and Manufacturing**, Springer, 2019.
2. Boccato C, Cerutti S, Vienken J, editors, **Medical devices: improving health care through a multidisciplinary approach** [Internet]. Springer International Publishing, 2022

References

1. Sokovic and Kopac, **RE as necessary phase by rapid product development**, Journal of Materials Processing Technology, 2005
2. Eldad Eilam, **Reversing: Secrets of Reverse Engineering**, Wiley, 2005.
3. Chris Eagle, **The IDA Pro Book: The Unofficial Guide to the World's Most Popular Disassembler**.